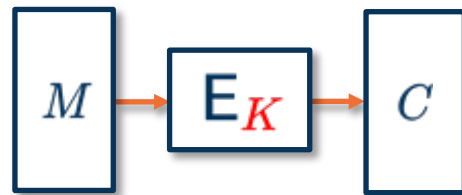


BLOCKCIPHERS AND THEIR SECURITY

Blockciphers, the Main Tool of Symmetric Crypto

◆ Examples: DES, 3DES, AES...

- A tool to encrypt short strings
- A block cipher is a function family $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$, where k (key length), n (input and output lengths) are the parameters



- Notation: for every $K \in \{0, 1\}^k$, $E_K : E(K, M)$
- For every $K \in \{0, 1\}^k$, $E_K(\cdot)$ is a permutation (one-to-one and onto function) on $\{0, 1\}^n$
For every $C \in \{0, 1\}^n$ there is a single $M \in \{0, 1\}^n$ s.t. $C = E_K(M)$
- Each blockcipher has an inverse denoted $E_K^{-1}(\cdot)$ s.t. $E_K(E_K^{-1}(C)) = C, E_K^{-1}(E_K(M)) = M$ for all $M, C \in \{0, 1\}^n$

DES

- ◆ Key length $k=56$, input and output length $n=64$
- ◆ **1973.** NBS (National Bureau of Standards) announced a search for a data protection algorithm to be standardized
- ◆ **1974.** IBM submits a design based on “Lucifer” algorithm
- ◆ **1975.** The proposed DES is published
- ◆ **1976.** DES approved as a federal standard
- ◆ DES is highly efficient: $\approx 2.5 \cdot 10^7$ DES computations per second

Security of Blockciphers

- Any blockcipher (and any symmetric encryption scheme) E is subject to exhaustive key-search: given $(M_1, C_1 = E(K, M_1)), \dots, (M_q, C_q = E(K, M_q))$ an adversary can recover K (or another key consistent with the given pairs) as follows:

$EKS_E((M_1, C_1), \dots, (M_q, C_q))$

For $i = 1, \dots, 2^k$ do

if $E(T_i, M_1) = C_1$ then // T_i is i -th k -bit string//

if $E(T_i, M_j) = C_j$ for all $2 \leq j \leq q$ then return T_i EndIf

EndIf

EndFor

Security of Blockciphers

- ◆ Key length $k=56$, input and output length $n=64$
- ◆ Exhaustive key search takes 2^k blockcipher computations in the worst case.
- ◆ On the average it takes approx. takes 2^{k-1} computations.
- ◆ DES has a property that $\text{DES}_K(x) = \overline{\text{DES}_{\overline{K}}(\overline{x})}$, this speeds up exhaustive search by a factor of 2.
- ◆ For DES ($k=56$), exhaustive search takes $2^{55}/2 \cdot 2.5 \cdot 10^7$ that is about 23 years.

Security of DES

- Key length $k=56$, input and output length $n=64$
- There are more sophisticated attacks known:
 - differential cryptanalysis: finds the key given about 2^{47} **chosen** plaintexts and the corresponding ciphertexts
 - linear cryptanalysis: finds the key given about 2^{42} **known** plaintext and ciphertext pairs
- These attacks require too many data, hence exhaustive key search is the best known attack. And it can be mounted in parallel!
- A machine for DES exhaustive key search was built in 1998 for \$250,000. It can find the key in about 56 hours on average. In 1999, the faster attack was mounted: 22 hours.
- A new block cipher was needed....
- Triple-DES: $3DES(K_1 || K_2, M) = DES(K_2, DES^{-1}(K_1, DES(K_2, M)))$
 - 3DES's keys are 112-bit long. Good, but needs 3 DES computations

Advanced Encryption Standard (AES)

- ◆ **1998.** NIST announced a search for a new block cipher
 - 15 algorithms from different countries were submitted
- ◆ **2001.** NIST announces the winner: an algorithm Rijndael, designed by Joan Daemen and Vincent Rijmen from Belgium
 - AES: block length $n=128$, key length k is variable: 128, 192 or 256 bits
 - Exhaustive key search is believed infeasible

Limitations of Key-recovery based Security

- ◆ A classical approach to blockcipher security: key recovery should be infeasible.
- ◆ But this is not good enough. Why?
- ◆ What property of a blockcipher as a building block would ensure various security properties of different constructions?

Intuition

- ◆ We want that (informally)
 - key search is hard
 - a ciphertext does not leak bits of the plaintext
 - a ciphertext does not leak any function of a plaintexts
 -
 - There is a “master” property of a blockcipher as a building block that enables security analysis of protocols based on blockciphers
- ◆ Idea: We will say that a blockcipher is good if ciphertexts “look” random
- ◆ A bit more precisely: A blockcipher is good if no efficient attacker can’t tell it apart from a random function

- Let $\text{Func}(l, L)$ denote the set of all functions from $\{0, 1\}^l$ to $\{0, 1\}^L$
- We will often consider the case when $l = L$
- Let's try to understand how a random function (a random instance f of $\text{Func}(l, L)$) behaves

Random Functions

- $g(\cdot) \stackrel{\$}{\leftarrow} \text{Func}(l, L)$
- We are interested in the input-output behavior of a random function. Let's imagine that we have access to a subroutine that implements such a function:

$g(X)$

global array T

If $T[X]$ is not defined then

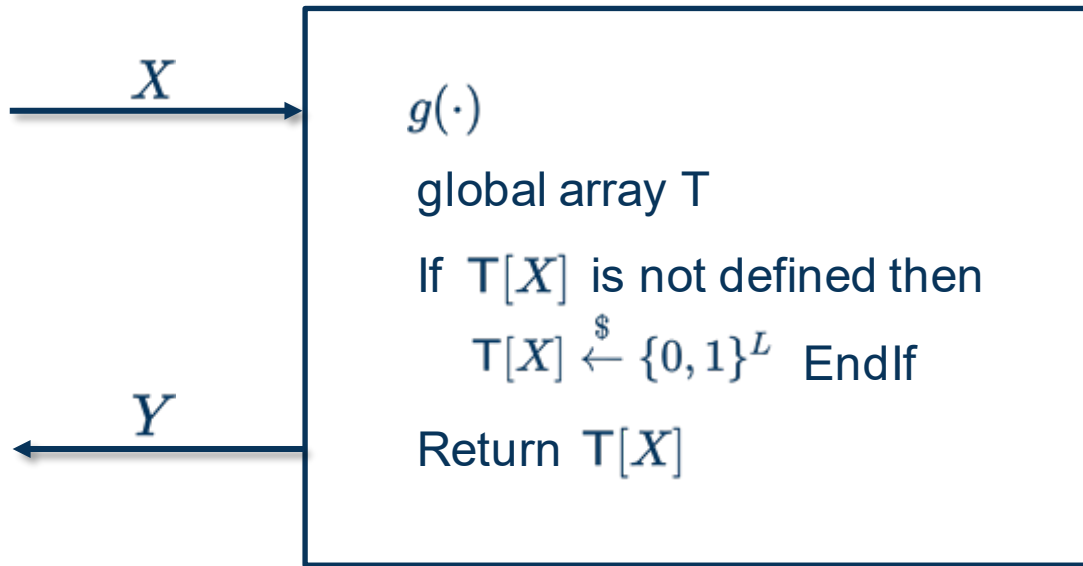
$T[X] \stackrel{\$}{\leftarrow} \{0, 1\}^L$ EndIf

Return $T[X]$

Pseudorandom Functions (PRFs)

- ◆ Informally, a function family F is a PRF if the input-output behavior (via “black-box” access) of its random instance is computationally indistinguishable from that of a random function.

“Black Box” Access

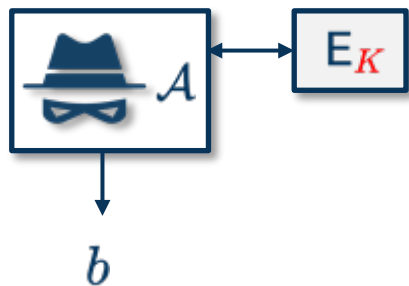


PRF Security for Blockciphers

- Def. Fix a blockcipher or function E . Consider the experiments:

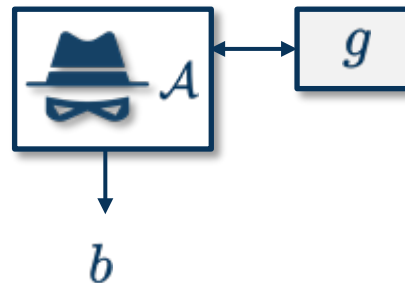
Experiment prf-1 (“real”)

Pick a random key K for E



Experiment prf-0 (“random”)

Pick a random function g from the set of all functions with the domain and range of E



The PRF advantage of $\mathcal{A}$ (denoted as $\text{Adv}^{\text{prf}}(\mathcal{A})$) is

$$\Pr[\mathcal{A} \Rightarrow 1 \text{ in Exp. prf-1}] - \Pr[\mathcal{A} \Rightarrow 1 \text{ in Exp. prf-0}].$$

E is a secure PRF if for any adversary with “reasonable” resources its prf advantage is “small”.

Resources of an adversary

1. Time-complexity is measured in some fixed RAM model of computation and includes the maximum of the running-times of $\mathcal{A}$ in the experiments, plus the size of the code for $\mathcal{A}$.
2. The number of queries $\mathcal{A}$ makes.
3. The total length of all queries.

Security of Practical Blockciphers

◆ Conjecture:

- AES is PRF secure.
- For any adversary $\mathcal{A}$ running time t and making q queries to its oracle,
$$\text{Adv}^{\text{prf}}(\mathcal{A}) \leq \text{const} \cdot t/T_{\text{AES}} \cdot 2^{-128} + q^2 \cdot 2^{-128}.$$